

Chapter 5A

Kidney Disease Overview: AKI, CKD, Urinalysis, and Renal Syndromes

Part V. Nephrology, Fluids, and Acid-Base

Chapter purpose

This starting renal-disease chapter builds a practical diagnostic framework for acute kidney injury, chronic kidney disease, urinalysis, proteinuria, hematuria, casts, renal imaging, initial management, and renal replacement therapy. It is designed as an original Medvale study chapter and future graph-RAG source: dense with mechanisms, numeric anchors, diagnostic relationships, and management links.

Core learning targets

- Separate AKI from CKD using time course, prior creatinine, kidney size, anemia/mineral abnormalities, urine findings, and reversibility.
- Classify AKI into pre-renal, intrinsic renal, and post-renal patterns before anchoring on any single lab value.
- Use urine sediment, urine sodium, FeNa, FeUrea, urine osmolality, and renal ultrasound as complementary clues rather than absolute rules.
- Recognize major CKD complications: hypertension, volume overload, anemia, metabolic acidosis, hyperkalemia, hyperphosphatemia, hypocalcemia, secondary hyperparathyroidism, bone disease, pruritus, infection risk, and uremic symptoms.
- Identify emergency indications for dialysis and distinguish urgent stabilization from long-term kidney-protective care.

Coverage signals used

The uploaded renal disease source emphasizes AKI definitions, pre-renal/intrinsic/post-renal categories, FeNa/FeUrea, urine sediment, renal ultrasound, CKD features, CKD complications, and dialysis indications. The source pages include original textbook diagrams/tables for causes of AKI, AKI evaluation, urinalysis findings, AKI versus CKD, CKD complications, and dialysis triggers. This Medvale chapter uses those topics as a coverage map while presenting original wording, organization, tables, and diagrams.

Clinical accuracy was cross-checked against KDIGO AKI and CKD guidance, including current CKD risk stratification by cause, GFR, and albuminuria and standard AKI diagnostic/staging concepts.

1. Why this matters clinically

Kidney disease is an internal medicine amplifier: a modest fall in filtration can rapidly change drug levels, potassium balance, acid-base status, volume status, blood pressure, endocrine signaling, bone-mineral metabolism, and prognosis. Acute kidney injury is often a clue to systemic instability - hemorrhage, sepsis, decompensated heart failure, obstruction, nephrotoxin exposure, rhabdomyolysis, vasculitis, or glomerulonephritis. Chronic kidney disease is both a disease state and a risk marker; low eGFR and albuminuria predict cardiovascular events, hospitalization, medication toxicity, and progression to kidney failure.

The key clinical move is to treat the kidney as a perfusion sensor, filter, endocrine organ, and urine-producing diagnostic window. Creatinine tells you that filtration has changed, but the urine tells you why. A bland concentrated urine with low urine sodium suggests intact tubules responding to low effective arterial blood volume; muddy granular casts suggest tubular injury; dysmorphic RBCs and RBC casts point toward glomerular inflammation; WBC casts suggest renal parenchymal inflammation such as pyelonephritis or interstitial nephritis. Renal ultrasound can change the problem from biochemical uncertainty to mechanical obstruction within minutes.

High-yield framing

Do not start with 'renal failure' as the diagnosis. Start with **time course** (acute vs chronic), **hemodynamics** (perfusion vs congestion), **urine sediment** (bland vs cellular vs granular), **protein/blood pattern**, **medication/toxin exposure**, and **obstruction screen**.

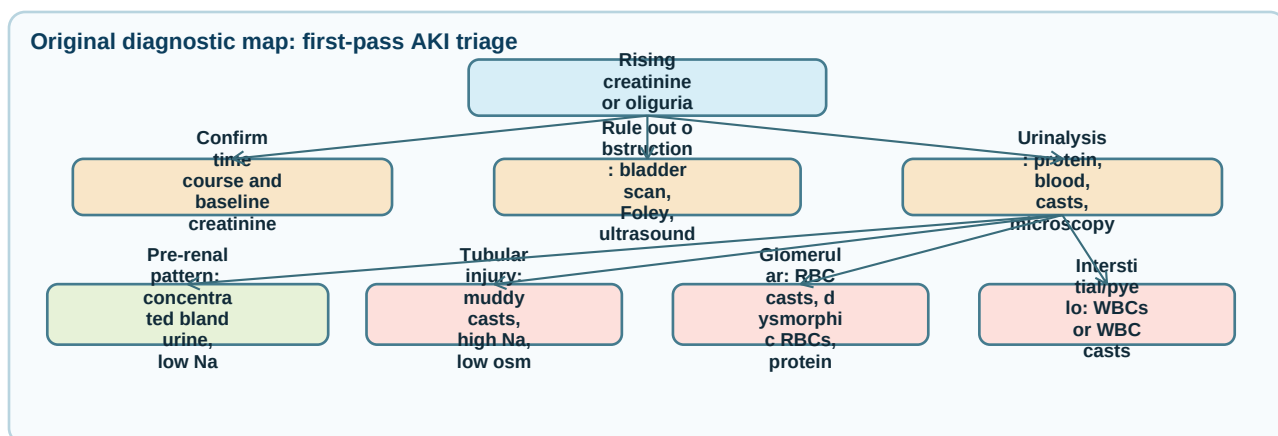
2. Core illness scripts

Syndrome	Typical trigger	Pathophysiologic center	Urine/lab clues	First management move
Pre-renal AKI	Vomiting, diarrhea, hemorrhage, overdiuresis, sepsis, cirrhosis, heart failure, NSAID/ACEi physiology	Low effective arterial blood volume lowers renal perfusion; tubules remain structurally intact and conserve sodium/water.	BUN:Cr often >20:1; urine osm often >500 mOsm/kg; urine Na often <20 mEq/L; FeNa <1% unless diuretics; bland/hyaline sediment.	Restore effective perfusion: fluids if volume-depleted; relieve congestion if cardiorenal; stop hemodynamic nephrotoxins.
Acute tubular injury/necrosis	Ischemia, sepsis, shock, pigment nephropathy, aminoglycosides, amphotericin, cisplatin, contrast, myoglobin	Tubular epithelial injury impairs sodium reabsorption, concentrating ability, and flow through tubules.	BUN:Cr often <20:1; urine osm often <350; urine Na >40; FeNa >2%; muddy brown granular casts/tubular cells.	Supportive care, remove cause, dose-adjust medications, manage K+/acid/volume, avoid repeated injury.
Post-renal AKI	BPH/prostate cancer, stones, tumors, retroperitoneal fibrosis, neurogenic bladder, catheter obstruction	Back pressure decreases net filtration; bilateral obstruction or solitary kidney obstruction is usually required for creatinine rise.	Hydronephrosis on ultrasound may appear; urine indices variable; bladder scan may show retention.	Decompress bladder/collecting system, monitor for post-obstructive diuresis and electrolyte shifts.
Glomerular syndrome	IgA nephropathy, post-infectious GN, lupus, ANCA vasculitis, anti-GBM disease	Inflamed filtration barrier leaks RBCs and protein; GFR may fall quickly.	Dysmorphic RBCs, RBC casts, proteinuria, hypertension, edema; complement/serology patterns guide workup.	Assess severity; nephrology; serologies; biopsy when needed; immunosuppression depends on diagnosis.
CKD	Diabetes, hypertension, chronic GN, polycystic kidney disease, reflux/obstruction, repeated AKI	Persistent structural or functional kidney abnormality for >3 months with health implications.	eGFR category G1-G5 plus albuminuria A1-A3; anemia/mineral-bone disease/acidosis/hyperK may appear as GFR declines.	Identify cause, slow progression, manage complications, reduce CV risk, prepare for kidney replacement if advanced.

3. Pathophysiology: what creatinine cannot tell you alone

Creatinine is a delayed and imperfect filtration marker. It rises only after filtration has fallen enough and time has passed for accumulation. A small absolute creatinine change can reflect a large GFR change in a patient with low baseline creatinine, and a 'normal' creatinine can conceal major kidney injury in patients with low muscle mass. Creatinine also changes with muscle breakdown, diet, tubular secretion, and medications that inhibit secretion. BUN is even less specific: it rises with low renal perfusion, catabolic states, gastrointestinal bleeding, corticosteroids, and high protein intake.

The kidney's response to injury depends on whether tubules remain intact. In low perfusion with intact tubules, the kidney avidly reabsorbs sodium and water under sympathetic, RAAS, and ADH influence. The urine becomes concentrated and sodium-poor. When tubules are damaged, they cannot conserve sodium or concentrate urine effectively; sodium wasting, lower urine osmolality, and granular casts appear. In obstruction, the kidney initially filters against back pressure, then GFR falls, tubular concentrating defects develop, and post-obstructive diuresis may occur after relief.



4. Diagnostic approach to AKI

AKI is diagnosed by change from baseline, not by a single creatinine threshold. Standard KDIGO criteria define AKI by any of the following: serum creatinine increase ≥ 0.3 mg/dL within 48 hours, increase to ≥ 1.5 times baseline known or presumed within 7 days, or urine volume < 0.5 mL/kg/hour for 6 hours. Staging is based on creatinine fold-change and duration/severity of oliguria; stage 3 includes creatinine ≥ 3 times baseline, creatinine ≥ 4.0 mg/dL with acute rise, initiation of kidney replacement therapy, or prolonged severe oliguria/anuria.

AKI stage	Creatinine criterion	Urine-output criterion	Clinical implication
Stage 1	≥ 0.3 mg/dL rise in 48 h OR 1.5-1.9x baseline	< 0.5 mL/kg/h for 6-12 h	Often reversible; search for hemodynamic, medication, and obstructive triggers.
Stage 2	2.0-2.9x baseline	< 0.5 mL/kg/h for ≥ 12 h	Higher complication risk; active electrolyte/volume surveillance.
Stage 3	3x baseline, creatinine ≥ 4.0 mg/dL with acute rise, or kidney replacement therapy	< 0.3 mL/kg/h for ≥ 24 h or anuria ≥ 12 h	High-risk AKI; consider ICU/nephrology; dialysis if complications.

The workup begins with four simultaneous questions: Is this truly acute? Is there an immediately reversible perfusion or medication cause? Is there obstruction? Does the urine sediment suggest intrinsic renal pathology? The minimum early dataset usually includes prior creatinine trend, intake/output and daily weight, vital signs and volume exam, medication review, urinalysis with microscopy, urine protein assessment, serum chemistries including potassium and bicarbonate, CBC, and renal/bladder imaging when obstruction is plausible or the cause is unclear.

Common AKI medication traps

NSAIDs constrict the afferent arteriole by blocking prostaglandin-mediated vasodilation; ACE inhibitors and ARBs dilate the efferent arteriole; diuretics can create pre-renal physiology; aminoglycosides, amphotericin, cisplatin, contrast, and pigment injury can damage tubules; PPIs, beta-lactams, NSAIDs, and other drugs can trigger acute interstitial nephritis.

5. Urinalysis and urine chemistries

Urinalysis is a structured biopsy of the nephron's output. Dipstick protein is most sensitive to albumin and can underestimate non-albumin proteins. Dipstick blood detects heme pigment, so a positive blood result with few or no RBCs suggests hemoglobinuria or myoglobinuria. Microscopy localizes injury: RBC casts imply bleeding from within the nephron, not from the ureter or bladder; WBC casts imply renal parenchymal inflammation; granular casts indicate cellular debris and tubular injury; hyaline casts may appear with concentrated urine and low-flow states.

Finding	Mechanistic meaning	Representative diagnoses	Pitfall
Hyaline casts	Concentrated Tamm-Horsfall protein matrix in low-flow urine	Pre-renal azotemia, dehydration, diuretic use, exercise	Not specific; interpret with urine concentration and clinical context.
Muddy brown granular casts	Tubular epithelial injury and debris	Acute tubular injury/necrosis from ischemia, sepsis, nephrotoxins, pigment injury	Can appear after pre-renal injury has progressed to ATN.
RBC casts or dysmorphic RBCs	Glomerular bleeding through inflamed filtration barrier	IgA nephropathy, lupus nephritis, ANCA GN, anti-GBM disease, post-infectious GN	Absence does not exclude GN if sample is dilute or delayed.
WBC casts	Inflammation within kidney tubules/interstitium	Pyelonephritis, acute interstitial nephritis	Urine eosinophils are neither sensitive nor specific for AIN.
Heavy albuminuria	Glomerular permeability problem	Diabetic kidney disease, membranous nephropathy, FSGS, minimal change disease	Dipstick can miss light chains; quantify with ACR/PCR and clinical suspicion.
Blood positive with few RBCs	Free heme pigment	Rhabdomyolysis, hemolysis	Check CK and clinical context; myoglobin can precipitate tubular injury.

Urine sodium, FeNa, and FeUrea estimate tubular sodium/urea handling, but they are not diagnostic in isolation. FeNa is often <1% in pre-renal physiology and >2% in tubular injury. FeUrea may be more useful when loop or thiazide diuretics elevate urine sodium, with values <35% supporting pre-renal physiology. However, contrast nephropathy, pigment injury, early obstruction, glomerulonephritis, and sepsis-associated AKI can violate simple thresholds. Treat urine indices as Bayesian evidence, not labels.

Index	Formula	Pattern favoring pre-renal	Pattern favoring tubular injury	Major limitations
FeNa	$(\text{Urine Na} \times \text{Plasma Cr}) / (\text{Plasma Na} \times \text{Urine Cr}) \times 100$	<1%	>2-3%	Diuretics, CKD, contrast, GN, sepsis, pigment injury, obstruction.
FeUrea	$(\text{Urine urea} \times \text{Plasma Cr}) / (\text{Plasma urea} \times \text{Urine Cr}) \times 100$	<35%	>50% often supports intrinsic injury	Less validated in some settings; affected by catabolism and sampling.
Urine osm	Measured mOsm/kg	>500 mOsm/kg if ADH/tubules intact	<350 mOsm/kg when concentrating ability impaired	May be high in glycosuria, mannitol, radiocontrast, or high solute excretion.
Urine Na	Spot urine sodium	<20 mEq/L when avid sodium retention	>40 mEq/L when tubular sodium wasting	Diuretics and CKD reduce specificity.

6. Pre-renal, intrinsic, and post-renal AKI

Pre-renal AKI is kidney hypoperfusion without primary structural damage. The causes include true volume loss, low cardiac output, systemic vasodilation, renal vasoconstriction, hepatorenal physiology, and medications that disrupt autoregulation. The exam may show dry mucosa and orthostasis in true depletion, but heart failure and cirrhosis can

produce low effective arterial blood volume despite edema and ascites. Improvement after restoring effective perfusion supports the diagnosis, but prolonged hypoperfusion can transition to acute tubular injury.

Intrinsic AKI means the kidney parenchyma is injured: tubules, glomeruli, vessels, or interstitium. Acute tubular injury is common after ischemia or nephrotoxins. Glomerular disease often presents with hematuria, RBC casts, proteinuria, hypertension, and reduced GFR. Vascular causes include renal artery occlusion, malignant hypertension, thrombotic microangiopathy, and cholesterol emboli. Interstitial nephritis often follows drug exposure and may have pyuria, WBC casts, rash, fever, eosinophilia, or none of the classic triad.

Post-renal AKI is obstruction to urine flow. Because one unobstructed kidney can maintain filtration, creatinine typically rises with bilateral obstruction, bladder outlet obstruction, or obstruction of a solitary functioning kidney. BPH is a common adult cause, but stones, malignancy, neurogenic bladder, urethral stricture, catheter obstruction, and retroperitoneal processes are important. Relief can cause brisk post-obstructive diuresis, requiring monitoring and replacement of water, sodium, potassium, magnesium, and phosphate losses.

Feature	Pre-renal	Acute tubular injury	Glomerular/interstitial	Post-renal
Clinical context	Volume loss, low EABV, shock, CHF/cirrhosis, NSAID/ACEi/ARB physiology	Prolonged ischemia, sepsis, toxins, pigment, contrast	Systemic symptoms, autoimmune disease, infection, drug exposure	BPH, stones, malignancy, neurogenic bladder, catheter problem
Sediment	Bland; hyaline casts	Granular casts, tubular epithelial cells	RBC casts in GN; WBC casts in pyelo/AIN	Variable; may be bland, RBCs, WBCs
Urine sodium/FeNa	Low unless diuretics/CKD	Higher due to tubular dysfunction	Variable	Variable by timing
Imaging	No hydronephrosis	Usually no hydronephrosis	Usually no hydronephrosis	Hydronephrosis or bladder retention may be present
Key first move	Restore effective perfusion	Remove insult and support	Serologies/biopsy/nephrology as needed	Decompress obstruction

7. Initial management of AKI

Management begins before the final etiology is known. Stop or hold nephrotoxins and renally cleared high-risk drugs, review contrast exposure, check volume status repeatedly, trend creatinine and urine output, dose-adjust medications to current kidney function, and treat dangerous complications immediately. If volume depleted, balanced crystalloid or isotonic saline can restore perfusion; if congested, diuretics and afterload/heart failure management may improve renal venous congestion and effective perfusion. The wrong fluid strategy can worsen AKI: repeated boluses in a congested oliguric patient may produce pulmonary edema, whereas aggressive diuresis in true hypovolemia can worsen ischemia.

Dialysis is initiated for complications rather than for a creatinine number alone. The classic emergency framework is AEIOU: refractory severe Acidosis, dangerous Electrolyte disturbance especially hyperkalemia, selected Intoxications, refractory Overload, and Uremic complications such as pericarditis, encephalopathy, seizures, or severe bleeding. In practice, clinical trajectory matters: a rapidly rising potassium in anuric AKI is more urgent than the same creatinine in a stable recovering patient.

Dialysis trigger	Examples	Why it matters
Acidosis	Severe metabolic acidosis not responding to medical therapy	Low pH depresses cardiac function and worsens hyperkalemia.
Electrolytes	Refractory hyperkalemia; sometimes severe hypermagnesemia or hyperphosphatemia	Hyperkalemia can cause fatal arrhythmia; ECG may be insensitive.
Intoxications	Toxic alcohols, lithium, salicylates, and other dialyzable toxins depending on level and symptoms	Dialysis removes toxin and corrects associated acid-base/electrolyte problems.
Overload	Pulmonary edema or severe volume overload unresponsive to diuretics	Respiratory failure and hypertension may become life-threatening.
Uremia	Pericarditis, encephalopathy, seizures, platelet dysfunction/bleeding, severe nausea/vomiting	Symptoms indicate clinically important toxin accumulation.

AKI monitoring bundle

Daily weight, strict intake/output, urine output trend, BMP with K⁺/HCO₃⁻/phosphate as needed, medication renal dosing, review for nephrotoxins, volume reassessment, urinalysis/microscopy, and obstruction screen when not clearly excluded.

8. Chronic kidney disease: definition and staging

Chronic kidney disease is defined by abnormalities of kidney structure or function present for more than 3 months with health implications. Reduced eGFR below 60 mL/min/1.73 m² for at least 3 months qualifies, but CKD can also exist with preserved eGFR when there are markers of kidney damage such as albuminuria, urine sediment abnormalities, imaging abnormalities, tubular disorders, pathologic abnormalities, or history of kidney transplantation. Modern CKD classification uses CGA: Cause, GFR category, and Albuminuria category.

This matters because albuminuria is not just a kidney clue; it is a vascular risk marker. A patient with eGFR 70 but ACR >300 mg/g has a high-risk kidney phenotype, while an older patient with eGFR 58 and no albuminuria may have lower short-term progression risk. Staging should therefore report both G and A categories, not eGFR alone.

GFR category	eGFR (mL/min/1.73 m ²)	Interpretive label
G1	≥90	Normal or high; CKD only if other kidney-damage marker present
G2	60-89	Mildly decreased; CKD only if other marker present
G3a	45-59	Mildly to moderately decreased
G3b	30-44	Moderately to severely decreased
G4	15-29	Severely decreased
G5	<15	Kidney failure range; evaluate for kidney replacement planning if appropriate

Albuminuria category	Urine ACR	Interpretive label
A1	<30 mg/g	Normal to mildly increased
A2	30-300 mg/g	Moderately increased albuminuria
A3	>300 mg/g	Severely increased albuminuria

Common causes include diabetes, hypertension, chronic glomerular disease, polycystic kidney disease, chronic obstruction/reflux, and repeated or severe AKI. Clues favoring chronicity include long-standing hypertension, diabetes, known abnormal urinalysis, small echogenic kidneys, anemia, hyperphosphatemia, hypocalcemia, secondary hyperparathyroidism, and kidney function that fails to normalize. However, normal-sized kidneys do not exclude CKD: diabetic kidney disease, amyloidosis, HIV-associated nephropathy, myeloma-related kidney disease, and polycystic kidney disease can have normal or enlarged kidneys.

9. Distinguishing AKI, CKD, and acute-on-chronic kidney disease

The most reliable discriminator is prior data. A creatinine of 2.0 mg/dL can represent a new emergency in one patient and stable CKD in another. When prior labs are unavailable, use indirect clues: chronic anemia, mineral-bone disease, small echogenic kidneys, long-standing albuminuria, neuropathy/retinopathy in diabetes, or a history of hypertension and abnormal urinalysis favor CKD. Rapid oliguria, abrupt exposure to shock/toxin/obstruction, active sediment, or recent normal creatinine favors AKI. Many hospitalized patients have acute-on-chronic disease: a chronic reduced nephron reserve with a superimposed hemodynamic or toxic injury.

Clue	Favors AKI	Favors CKD	Caveat
Prior creatinine	Recent normal value or abrupt rise	Abnormal for >3 months	No prior data is common; avoid assuming chronicity.

Clue	Favors AKI	Favors CKD	Caveat
Kidney ultrasound	Normal size possible	Small echogenic kidneys suggest chronic irreversible disease	Diabetes, amyloid, HIV, myeloma, and PKD may have large/normal kidneys.
Urine output	Oliguria/anuria can be abrupt	May remain normal until advanced CKD	Obstruction can cause anuria and is reversible if treated early.
Sediment	New granular/cellular casts	May be bland or chronically proteinuric	Active GN can be acute or chronic flare.
Complications	Hyperkalemia/acidosis can develop suddenly	Anemia, bone-mineral disease, pruritus, neuropathy suggest chronicity	Severe AKI can also cause hyperphosphatemia/acidosis.

10. CKD complications and management links

CKD complications reflect loss of filtration, endocrine function, tubular regulation, and immune competence. Sodium and water retention contribute to hypertension and edema. Reduced erythropoietin production causes typically normocytic anemia, though iron deficiency, inflammation, blood loss, and malnutrition frequently coexist. Reduced ammoniogenesis and nephron mass contribute to non-anion-gap or mixed metabolic acidosis. Impaired potassium excretion predisposes to hyperkalemia, especially with RAAS blockade, hypoaldosteronism, acidosis, or high dietary potassium. Reduced phosphate excretion and reduced calcitriol production cause hyperphosphatemia, hypocalcemia, secondary hyperparathyroidism, renal osteodystrophy, vascular calcification, and calciphylaxis risk in severe disease.

Long-term management has two parallel goals: slow progression and reduce complications. Slowing progression centers on treating the cause, blood pressure control, albuminuria reduction, diabetes optimization, smoking cessation, avoidance of nephrotoxins, cardiovascular risk reduction, and careful medication dosing. Complication management includes diuretics for volume overload, bicarbonate for clinically relevant metabolic acidosis when appropriate, phosphate control, vitamin D/calcitriol strategies in selected patients, iron/ESA pathways for anemia, vaccination/infection prevention, and timely education about dialysis, transplant, and conservative care options.

Complication	Mechanism	Clinical/lab clue	Management link
Hypertension/edema	Sodium retention, RAAS activation, vascular stiffness	Rising BP, weight gain, edema, pulmonary congestion	Salt restriction, diuretics, BP regimen, assess volume.
Anemia	Low erythropoietin plus iron/inflammation contributors	Normocytic anemia, fatigue, dyspnea	Check iron indices; consider ESA only in appropriate CKD context.
Metabolic acidosis	Reduced acid excretion/ammoniogenesis	Low serum bicarbonate	Treat cause; oral alkali in selected CKD patients.
Hyperkalemia	Reduced K ⁺ excretion, RAAS blockade, acidosis, hypoaldosteronism	K ⁺ elevation, ECG may be normal	Stabilize if severe; shift/remove K ⁺ ; review diet/medications.
CKD-MBD	Phosphate retention, low calcitriol, high PTH	High phosphate, low/normal calcium, high PTH, bone pain/fracture risk	Phosphate control, vitamin D strategies, PTH management.
Uremic symptoms	Accumulation of retained solutes	Pericarditis, encephalopathy, nausea, pruritus, bleeding	Evaluate for kidney replacement therapy if severe.

11. Renal syndrome recognition

A renal syndrome is a pattern that narrows the differential before the exact disease is known. Nephrotic syndrome is dominated by heavy proteinuria, hypoalbuminemia, edema, hyperlipidemia, lipiduria, and thrombosis risk. Nephritic syndrome is dominated by inflammatory hematuria, RBC casts, hypertension, reduced GFR, and variable proteinuria. Tubulointerstitial syndromes often have pyuria, WBC casts, tubular dysfunction, mild-to-moderate proteinuria, and medication or infection associations. Obstructive syndromes may present with oliguria, hydronephrosis, flank pain, retention, recurrent infections, or post-obstructive diuresis after relief.

Syndrome	Dominant urine pattern	Dominant clinical pattern	Examples
Nephrotic	Protein often >3.5 g/day, lipiduria, bland or fatty casts	Edema, hypoalbuminemia, hyperlipidemia, thrombosis/infection risk	Minimal change, FSGS, membranous, diabetic kidney disease, amyloid.
Nephritic	Hematuria, dysmorphic RBCs, RBC casts, subnephrotic or mixed proteinuria	Hypertension, edema, reduced GFR, systemic inflammation possible	IgA, post-infectious GN, lupus, ANCA, anti-GBM.
Tubular injury	Granular casts, tubular cells, impaired concentration	AKI after ischemia/toxin/sepsis/pigment	ATN/acute tubular injury.
Interstitial	Pyuria, WBC casts, mild proteinuria; eosinophilia sometimes	Drug exposure, rash/fever may be absent, flank discomfort possible	AIN, pyelonephritis.
Obstructive	Variable sediment; hematuria/infection possible	Retention, hydronephrosis, flank pain, anuria in bilateral obstruction	BPH, stones, cancer, neurogenic bladder.

12. Differentials and mimics

Not every creatinine rise is kidney parenchymal damage. Pseudo-AKI occurs when creatinine generation increases or secretion decreases without true GFR fall. Trimethoprim and some other drugs inhibit tubular creatinine secretion. High meat intake or creatine supplements may raise creatinine. Rhabdomyolysis raises creatinine and simultaneously causes true tubular injury, so CK and urinalysis are needed. BUN disproportion may reflect GI bleeding, catabolism, steroids, or protein load rather than pre-renal physiology alone.

Oliguria can be physiologic conservation, catheter obstruction, bladder retention, shock, or intrinsic kidney injury. Edema can occur in nephrotic syndrome, heart failure, cirrhosis, venous disease, lymphatic obstruction, and medication effects. Hematuria can originate anywhere from glomerulus to urethra; dysmorphic RBCs and casts move the source into the nephron. Proteinuria can be albumin-predominant glomerular proteinuria, tubular proteinuria, overflow proteinuria from monoclonal light chains, or transient functional proteinuria from fever/exercise.

13. High-yield exam clues

- BUN:Cr >20:1, urine Na <20, FeNa <1%, urine osm >500, and bland/hyaline sediment: pre-renal physiology, but diuretics/CKD can distort urine sodium.
- Muddy brown granular casts after shock/sepsis/nephrotoxin exposure: acute tubular injury.
- RBC casts plus hypertension and reduced GFR: nephritic/glomerulonephritis pattern.
- Positive dipstick blood with few RBCs and very high CK: rhabdomyolysis with myoglobinuria.
- Older man with oliguria, suprapubic fullness, hydronephrosis, or large post-void residual: post-renal AKI until proven otherwise.
- Small echogenic kidneys, anemia, hyperphosphatemia, hypocalcemia, and symptoms over months: CKD.
- Dialysis is for refractory complications and uremic manifestations, not an isolated creatinine number.

14. Common pitfalls

- Assuming edema means total-body sodium excess is the only issue; heart failure and cirrhosis can have low effective arterial blood volume despite edema.
- Using FeNa after diuretics as if it were definitive; use FeUrea and clinical context instead.
- Missing obstruction because the patient still makes some urine; partial obstruction can be non-anuric.
- Giving repeated fluid boluses to oliguric cardiorenal syndrome without reassessing congestion.
- Failing to dose-adjust renally cleared medications during rapidly changing AKI.
- Calling CKD irreversible without looking for treatable superimposed AKI.

- Waiting for ECG changes before treating severe hyperkalemia; ECG sensitivity is imperfect.

15. Practice questions

Question 1. A 72-year-old man with vomiting has creatinine 2.1 from baseline 0.9, BUN 52, urine osm 650, urine sodium 8, FeNa 0.4%, and hyaline casts. What is the most likely AKI pattern?

Answer. Pre-renal AKI from volume depletion. The kidney is conserving sodium and water, and the sediment is bland.

Question 2. A septic ICU patient has rising creatinine after 2 days of hypotension. Urine sodium is 55, urine osm 280, FeNa 3.5%, and microscopy shows muddy granular casts. What changed pathophysiologically?

Answer. Prolonged hypoperfusion has injured tubules, converting a potentially pre-renal state into acute tubular injury with impaired sodium reabsorption and concentrating ability.

Question 3. A patient has dipstick blood 3+ but only 0-2 RBC/hpf after being found down for 10 hours. What test should be ordered and why?

Answer. Creatine kinase. The pattern suggests myoglobinuria from rhabdomyolysis, which can cause pigment-associated tubular injury.

Question 4. A 68-year-old man with BPH has anuria, suprapubic fullness, and creatinine 4.0. What is the immediate diagnostic/therapeutic step?

Answer. Bladder scan and urethral catheterization if retention is present, followed by monitoring for post-obstructive diuresis.

Question 5. A patient has eGFR 72 for 1 year and urine ACR 650 mg/g. Does this patient have CKD?

Answer. Yes. CKD can be present with preserved eGFR if kidney-damage markers persist for >3 months; this is G2A3 if eGFR 60-89 and ACR >300 mg/g.

Question 6. A CKD patient has confusion, pericardial friction rub, K⁺ 6.6 despite temporizing therapy, and pulmonary edema refractory to diuretics. What framework supports dialysis?

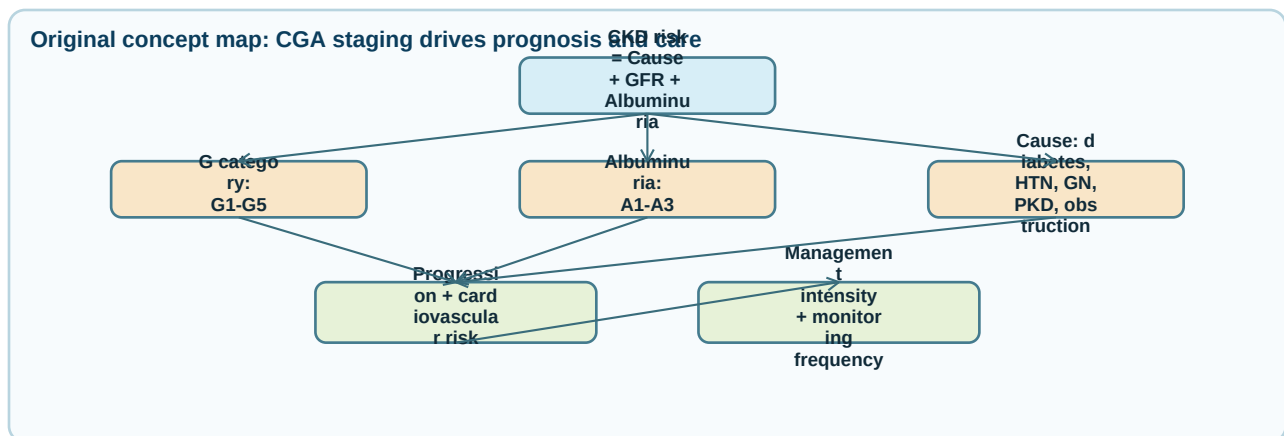
Answer. AEIOU: uremic complication, refractory electrolyte disturbance, and refractory overload are dialysis indications.

16. Graph-RAG extraction map

The map below is intended to help a future graph-RAG pipeline extract stable nodes, edges, evidence relationships, and clinical roles from this chapter.

Node	Class	Key attributes / facts	Suggested edges
Acute kidney injury	Renal syndrome	Creatinine rise ≥ 0.3 mg/dL/48 h, ≥ 1.5 x baseline/7 d, or oliguria < 0.5 mL/kg/h/6 h.	classified_by -> pre-renal/intrinsic/post-renal; causes -> electrolyte/acid-base complications
Pre-renal AKI	AKI subtype	Low effective arterial blood volume with intact tubules; low FeNa, high urine osm, bland sediment.	progresses_to -> acute tubular injury if prolonged; treated_by -> restoring effective perfusion
Acute tubular injury	Intrinsic AKI subtype	Tubular epithelial injury; muddy granular casts; impaired sodium/water handling.	caused_by -> ischemia/nephrotoxins/pigment; managed_by -> supportive care
Post-renal AKI	Obstructive syndrome	Urinary outflow obstruction; bilateral/solitary kidney obstruction usually needed for creatinine rise.	diagnosed_by -> bladder scan/ultrasound; treated_by -> decompression
Urinalysis microscopy	Diagnostic test	Casts and cells localize kidney injury pattern.	detects -> RBC casts/WBC casts/granular casts/hyaline casts
FeNa	Diagnostic index	Fractional excretion of sodium; $< 1\%$ often pre-renal, $> 2\%$ often tubular injury.	limited_by -> diuretics/CKD/sepsis/contrast/GN
FeUrea	Diagnostic index	Can support pre-renal physiology when diuretics confound FeNa; $< 35\%$ often pre-renal.	used_when -> diuretics present
Chronic kidney disease	Chronic renal syndrome	Kidney abnormality > 3 months; staged by cause, GFR, albuminuria.	complicated_by -> anemia/CKD-MBD/acidosis/hyperkalemia/volume overload
Albuminuria	Risk marker	A1 < 30 mg/g, A2 30-300, A3 > 300 .	risk_stratifies -> CKD; suggests -> glomerular injury
Dialysis indication	Management trigger	AEIOU: acidosis, electrolytes, intoxication, overload, uremia.	treats -> refractory AKI/CKD complications

Relationship classes to encode: **causes**, **progresses_to**, **classified_by**, **diagnosed_by**, **suggests**, **limited_by**, **treated_by**, **complicated_by**, **risk_stratifies**, and **contraindicates/cautions**.



17. References used for clinical cross-checking

KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney International. 2024.

KDIGO Clinical Practice Guideline for Acute Kidney Injury. Kidney International Supplements. 2012.

Medvale uploaded renal disease source file, used as a topic coverage guide for AKI, CKD, urinalysis, and dialysis sections.

Originality note: this chapter was newly organized and written for Medvale. Uploaded renal-source pages and Anki cards were used as topic-coverage signals only; tables, diagrams, explanations, practice questions, and graph-RAG mappings are original study-resource content.